

Two-pan weight kits

The mathematics in one view

The target stays on the left pan. Each positive whole-number weight is available once and can be placed on the left, on the right, or off the balance. A target t is possible exactly when $t = \sum \epsilon_i w_i$ with ϵ_i in $\{-1, 0, 1\}$; a positive sign means opposite the target, a negative sign means beside it.

Balanced ternary. The kit $1, 3, 9, \dots, 3^{m-1}$ balances every integer target from 1 through $(3^m - 1)/2$, in exactly one placement with the target fixed. For two, three, four and five weights, the endpoints are 4, 13, 40, 121.

Optimality. No m -weight kit can balance more than $(3^m - 1)/2$ distinct positive targets, even if gaps are allowed. There are 3^m placement choices; all-off gives zero, and reversing the two pans pairs each positive difference with a negative one. Collisions and additional zero differences only reduce the count. This counts placements in one exact balance, not outcomes of successive weighings.

Why powers of three fit. If the smaller weights have total S and represent every integer from $-S$ through S uniquely, adding $2S+1$ gives three disjoint adjacent intervals and unique coverage from $-(3S+1)$ through $3S+1$. A heavier new weight leaves a gap; a lighter one cannot reach as far.

What each group is doing

K-1: Move weights, compare small kits, and discover that adding beside the target can help. Count totals to 6 initially; the last problem reaches 13 with adult support. Negative numbers and equations are unnecessary.

Grades 2-3: Compare overlap, gap and exact fit for third weights 8, 9, 10; formulate the next-weight rule. Addition to 13, then a way to reason about intervals to 40, is helpful. The placement-count bound can be a shared adult-led explanation.

Grades 4-5: Explain uniqueness and the universal counting bound, distinguish many targets from an unbroken run, and extend to five weights. Powers can be counted as repeated groups of three; algebra is optional.

Materials and preparation

Five pair-kits of paper weight cards 1, 2, 3, 4, 8, 9, 10, 27, 81, five paper two-pan mats, pencils, and counters for checking small totals. Give only the current kit: two cards labeled 1 would violate the rules. Paper cards stand for amounts, not their actual mass. One real equal-arm balance is optional; use verified unit-cube bundles if demonstrating it. Do not expect paper disks of equal mass to behave like their labels.

Print the chosen entry page for each child and later pages on demand. All current adaptations are unpiloted.

Launch and a flexible hour

0-6 minutes: Let pairs move cards freely, then show the printed example: target 3 with weights 1 and 4. Put 1 beside the target and 4 opposite, so $3+1=4$. Leave an unused card off. Ask a child to explain why the totals agree.

6-23: K P1-3; 2-3 P1-3; upper P1-2. **23-28:** Stand and trade a target challenge with another pair.

28-48: Continue kit comparisons or move to duplicate balances, uniqueness and the counting bound.

48-60: Show one useful balance and one reason a kit must fail. Stop at small examples if extending the kit would replace understanding with arithmetic.

Small-kit key: shared across levels

Kit	All positive targets it balances	Reason
1,2	1,2,3	Single weights, sum 3, difference 1
1,3	1,2,3,4	$2+1=3$; also singles and sum 4
1,4	1,3,4,5	$3+1=4$; misses 2
1,3,8	1 through 12	Old differences $-4\dots4$, plus 8: $4\dots12$
1,3,9	1 through 13	Old differences $-4\dots4$, plus 9: $5\dots13$
1,3,10	$1\dots4$ and $6\dots14$	New positive interval starts at $10-4=6$

K-1 answer key

P1. $1=1$, $2+1=3$, $3=3$, $4=1+3$. The target is the first number on each left side; each listed weight is used once.

P2. Exactly 1,2,3 work with 1 and 2; 4 and 5 exceed their total weight 3.

P3. A tie in the *number* of targets: each kit reaches four of 1-6. The 1/4 kit reaches 1,3,4,5; the 1/3 kit reaches 1,2,3,4. Only the latter has no gap in its run.

P4. Only {1,3}. All six kits give: $\{1,2\} \rightarrow \{1,2,3\}$; $\{1,3\} \rightarrow \{1,2,3,4\}$; $\{1,4\} \rightarrow \{1,3,4,5\}$; $\{2,3\} \rightarrow \{1,2,3,5\}$; $\{2,4\} \rightarrow \{2,4,6\}$; $\{3,4\} \rightarrow \{1,3,4,7\}$. Every other kit misses a requested target.

P5. Choose 9. The 8-kit cannot reach 13 because its total is 12; the 10-kit cannot reach 5 because even putting both small weights beside 5 leaves $5+4<10$. The 9-kit constructions are on the next page.

Grades 2-3 answer key

P1. Use the small-kit table on page 2. A larger maximum is not necessarily a longer gap-free run.

P2. Choose 9. With a new weight w , reaching 13 requires $w+4 \geq 13$, hence $w \geq 9$. Reaching 5 requires $w-4 \leq 5$, hence $w \leq 9$; a small kit without w reaches at most 4. Thus 9 is the only choice.

Target	Balanced pans	Target	Balanced pans
1	$1 = 1$	8	$8+1 = 9$
2	$2+1 = 3$	9	$9 = 9$
3	$3 = 3$	10	$10 = 9+1$
4	$4 = 3+1$	11	$11+1 = 9+3$
5	$5+3+1 = 9$	12	$12 = 9+3$
6	$6+3 = 9$	13	$13 = 9+3+1$
7	$7+3 = 9+1$		

In this table the first number on the left is the target, not another weight. Unmentioned weights stay off.

P3. First missed positive targets: $1/3/8 \rightarrow 13$; $1/3/9 \rightarrow 14$; $1/3/10 \rightarrow 5$. The complete ranges on page 2 establish that these are the *first* misses.

P4. With 1,3,8 there are exactly two balances of 4: $4=3+1$ (8 off), and $4+3+1=8$. With 1,3,9 there is exactly one: $4=3+1$ (9 off). If the big weight is opposite the target, the small weights must supply the remaining difference; if it is beside the target, the opposite small weights cannot outweigh it. These cases exhaust the placements.

P5. No. Three weights have 27 placements, at most 13 distinct positive differences after zero and sign reversal are accounted for. Fourteen different positive targets require at least 29 placements.

P6. Add 27. The old kit gives every signed difference from -13 through 13. Put 27 opposite and use the old kit to adjust by any of these differences: this covers 14-40. Together with 1-13 it covers 1-40. A fourth weight below 27 has total below 40; above 27 it misses 14.

Hints, only when needed

For a stuck balance, ask whether a weight may sit beside the target. For comparing new weights, find the smallest target using the new weight and the largest using it. For “all ways,” split into big weight off / opposite / beside, rather than move cards randomly.

Grades 4-5 answer key and proofs

- P1.** Only $\{1,3\}$. For $0 < a < b$ the possible positive targets are among a , b , $b-a$, $a+b$. To cover 1,2,3,4, these four values must be distinct and equal that set. Their maximum $a+b$ must therefore be 4; the only distinct positive integer pair with that sum is 1,3.
- P2.** Add 9 to cover 1-13. If $w < 9$, total $w+4 < 13$. If $w > 9$, 5 is impossible: w opposite gives at least $w-4 > 5$; w beside cannot produce a positive target, and omitting w reaches at most 4.
- P3.** No duplicate placements exist for any positive target 1-13. Suppose two placements gave the same difference. Subtract them and take the largest weight where they differ. Its contribution has magnitude at least that weight; all smaller changes together have magnitude at most twice their sum. For 9 this is $2(1+3)=8 < 9$; for 3 it is $2 < 3$; for 1 it is $0 < 1$. Cancellation is impossible.
- P4.** No, by the 27-placement / 13-positive-target bound (page 3). It applies to any three weights, regardless of their magnitudes or whether some placements give zero.
- P5.** Counts depend on the chosen kit, but the maximum is 13. For example, 1,2,3 reaches only 1-6. The kit 1,3,9 attains 13 with no gaps. The kit 1,4,16 also attains 13 but reaches $\{1,3,4,5,11,12,13,15,16,17,19,20,21\}$. A kit cannot exceed 13 merely by allowing gaps.
- P6.** Add 27: the signed intervals are $[-40,-14]$, $[-13,13]$, $[14,40]$. They are disjoint and adjacent, with a unique old-kit placement inside each interval, so the new kit has neither gaps nor duplicates. Alternatively $27 > 2(1+3+9)=26$ proves uniqueness. The 81 placements bound the positive run by 40, so it is optimal.
- P7.** The maximum is $(3^5-1)/2=121$, achieved by 1,3,9,27,81. The old signed interval is $[-40,40]$; adding 81 gives $[-121,-41]$, $[-40,40]$, $[41,121]$, each uniquely represented. The 243 placement choices permit at most 121 positive targets.

Optional extension

If the group owns the argument, ask what changes when weights may only go opposite the target. Each has two choices; 1,2,4 cover 1-7 uniquely. This contrasts two versus three placement choices without introducing a different weighing puzzle.

Primary source

Ethan D. Bolker, Samuel A. Feuer and Catalin Zara, "Balance weighing - variations on a theme," *Mathematics Magazine* 94(5), 2021, author manuscript pp. 1-2, Theorem 1. It reports fourth graders exploring the two-pan sequence and states balanced-ternary uniqueness. Current finite cases and the placement-count proof were checked independently. [Read the author-hosted paper.](#)